CST463 - Advanced Machine Learning

CST463 (Advanced Machine Learning) Syllabus

Summary

In this course you will learn to use advanced machine learning methods, including dimensionality reduction, ensemble methods, neural nets, and methods for working with time series data.

- **Grading:** To pass this course you must (1) earn an overall grade of at least 60%, **and** (2) score at least 40% in **each** of the three assignment groups. They are Programming Assignments (40%), Exams (40%), and Participation (20%). Final grades are calculated by rounding to the nearest whole number and converted to letter grades using the standard range. *Details can be found in grading.*
- **Late work:** Late submissions are accepted only for Programming assignments and Labs, and have a 10% deduction per calendar day late. *Details can be found in late policy.*
- **Getting help:** There are a ton of office hours spread throughout the week so make use of them if you have any questions! In general, you can use the class slack channel to ask questions (but not post code!), or send me questions (code okay!) via slack or my email. Anything you submit, you should be able to explain and discuss — see the Academic Honor Code for the full policy and why. *Details on office hours can be found in personnel.*

Course Information

Course Details

- **Course Title:** Advanced Machine Learning
- **Course Number:** CST463
- **Prerequisites:** CST383

Course Description

In this course you will learn to use advanced machine learning methods, including dimensionality reduction, ensemble methods, neural nets, and methods for working with time series data.

Course Objectives

At the end of this class, you should be able to: - Apply deep learning to machine-learning problems of moderate complexity, using Google's TensorFlow library and Python. - Explain and implement machine learning algorithms using loss functions and optimization, including the use of backpropagation in neural nets. - Explain the principles behind dimensionality reduction and ensemble methods (including random forests and boosting). - Apply machine learning to time series data sets.

Topics Covered

The major topics covered in class are: 1. Working with Data & Math 2. Gradient Descent 3. Neural Network Basics 4. TensorFlow 5. Vision Models 6. Time Series 7. Text Models

Personnel

- Dr. Sam Ogden (instructor)
  - E-mail: `sogden@csumb.edu`
  - Web: **`https://csumb.edu/scd/sogden/`**
  - Office: BIT205
  - Office Hours: 10-11am Tuesdays & 1pm-2pm Wednesdays, or by appointment
- Achsah Jojo
  - Email: ajojo@csumb.edu
  - Office Hours: TBD

Getting Help: Office hours are available throughout the week - make use of them! You can also use slack to ask questions (but not post code/answers on public channels!), or send me questions via email.

Communicating online I will use email, Canvas, and Slack for online class communication.

1. **email** is to be used when you need to officially communicate something to me.
   - e.g. "Can you double check this assignment grade for me?"
2. **Slack** will be used to answer content questions and give quick updates.
   - e.g. "I think I found a bug in this quiz, could you check it out?"
   - We have a slack channel on the CS-U-Monterey workspace that you should join.
3. **Canvas** will have information on assignments and due dates
   - In general I don't read things in canvas messages or comments. For questions or comments please use Slack or email instead.

Materials

- Course textbook available for free online: Dive into Deep Learning
- Course lecture recordings: CST463 on Panopto (Note: Previous semesters videos are great for extra reference but also might change so coming to class is incredibly important!)
- Canvas Course Page: CST463 on Canvas
- Syllabus: CST463 Syllabus

Grading

There are three groups of assignments, briefly described in Table 1. **To pass this course, you must meet both of the following requirements: (1) achieve an overall passing grade (60%), and (2) score at least 40% in each of the three assignment groups below. Failing to meet either requirement will result in failing the course.**

Assignment Kind	Value
Homework	36%
Exams	36%
Final Project	16%
Participation	12%

Table 1. Points available per assignment type.

Homework

Homework will consist of three parts: - Development of a library of functions that perform a set of tasks - Visualization and analysis using the functions developed during the first half - Reviews of your classmates work

The goal of these three parts is to give you experience in the three major aspects of data analysis and machine learning: development of new module and code, analysis of data, interpreting the work of others.

Assignments will generally span two weeks, with the first week focusing on the development of library functions and review of the previous week's analysis, and the second week will focus on your own analysis and visualization.

Therefore, there will be three parts to how you are assess: 1. The python code you develop will be checked against unit tests to ensure expected functionality (roughly 50%) 2. Your visualization and analysis, based on how well I and your peers understand your interpretation and the validity (roughly 30%) 3. Your analysis of your peer's visualization and analysis. (roughly 20%)

Exams

There will be three exams in this course worth each worth roughly 12% of your grade. Each one will cover a mixture of new and old material, with an emphasis on new material. During the exam you will be allowed to use a calculator but no other electronics, and have a single sheet of notes.

These exams are spaced roughly 5 weeks apart and can be seen on the course calendar.

Note: there is no final exam for this class, but we may have a project presentations period during this block.

Participation

In-class participation has three components:

1. (4%) Attendance: Show up to class on time every day
2. (4%) Labs: We are going to be doing a number of in-class activities, and other skill-building exercises.
3. (4%) Weekly Study Notes: reflect on the material you learned each week and use as notes for studying

In-class quizzes Attendance will be taken through short in-class quizzes or surveys. These surveys may either be based on readings from OSTEP (the appropriate chapters for a day can be found in the class calendar) or be surveys used to check attendance. Quizzes are open-book and open-note and will be completed on canvas.

While these quizzes cannot be made up, the lowest 4 quizzes will be dropped. Please note that these quizzes may not occur every class, and are more likely on days when attendance is low, as a way to encourage attendance.

Labs Throughout the course there will be several labs that are intended to be completed in class. These are designed to help you better understand key topics, get practice with calculations, and get experience with the working environments. Generally, these are designed to take 30 minutes and will happen every two weeks, with submissions due on **Sunday night**. Students are strongly encouraged to work in groups on the labs and use any available outside resources. Although in general I aim to provide sufficient information to complete the labs, students are encouraged to use any additional resources they find helpful to complete them since they often are only an introduction to the material.

Weekly Study Notes Each week students complete a Weekly Study Note as a habit-building exercise: a short, low-stakes checkpoint where you pause and consolidate your own understanding of that week's material, ideally building toward a personalized study guide you can return to later.

Weekly Study Notes are graded on completion rather than content quality or depth — the goal is building the habit of regular reflection, not producing a polished document.

Weekly Study Notes are graded with the help of an LLM. Aggregated and anonymized, they're a useful signal for me: they help me see which topics need more coverage or a different explanation in future classes, so the more genuinely yours the notes are, the more useful this feedback loop is for the class as a whole.

Final Project

The final project will consist of you and teammates exploring a project idea related to advanced machine learning. I will suggest a few possible projects, but you are encouraged to think up ideas throughout the semester!

Extra Credit Potential

Bug Hunting We all make mistakes – my github commit history is proof of that. Sometimes you catch big ones before I do, and if you let me know I might throw some extra credit your way. This is generally reserved for big things (e.g. I left out a bunch of files in the repo or some tools out of the docker image), but if you find yourself confused by something please reach out, maybe you found a novel bug! [1]

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below, with exceptions made based on the general grading requirements guidance. Note that $[90, 93)$ means "at least 90 but less than 93" (e.g. $90 \leq score < 93$) and that all grades will be rounded to the nearest whole number (e.g. a 92.1 will be rounded to a 92, but a 92.5 will be rounded to a 93). Additionally, note that there will be no curve.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Course Policies

Attendance

Attending lecture is strongly correlated with academic achievement[2], and thus is strongly encouraged. Attendance is tracked through in-class activities including, but not limited to, reading quizzes.

If you are sick, please do not come to class.
A number of attendance quizzes are dropped to accommodate things that come up in life.

Late Policy

With two exceptions, no late work will be accepted and no extensions will be granted. These two exceptions are programming components of homework assignments and lab assignments, both of which can be submitted with a late penalty at any point before the Sunday prior to final exam week. This late penalty will be a 10% reduction in maximum points per day, with a maximum reduction of 60%[3].

[1] I should note that it's not like there's a place for direct extra credit for bugs, but I generally find a way to tweak things a bit depending on how big a bug it is. Regardless I always appreciate feedback.

[2] Attendance and grade are strongly correlated (not just because of the attendance quizzes but they also factor in).

[3] This is done automatically through canvas and it "chops off" the points that you can't get. For example, if you turn in an assignment 1 day late and would have gotten a 95% on it you will get a 90% for it.

If you do fall behind in class, please set up a time to meet with me to discuss getting you back on track – my goal is to have you succeed in class and I want to find a way to make that happen.

Academic Honor Code

The core standard: anything you submit, you should be able to explain and discuss. If you can't explain why your code does what it does, why you chose one approach over another, or walk through it line by line if asked, that's treated as an academic honor code concern — regardless of where the code or idea originally came from.

Why this standard, and not a rule about tools or line counts: rules like "don't use more than N lines from an LLM" are easy to argue around and hard to enforce consistently. What actually matters is whether *you* understand what you're submitting. This standard is also directly enforceable — I or a TA can ask you to walk through your code in office hours, during grading, or on a paper exam, and "I'm not sure, I just used what worked" is not an acceptable answer.

Idea-sourcing vs. code-sourcing

Allowed — talking to other students, TAs, or LLMs for general or mechanical questions not tied to your specific assignment: - Clarification on topics covered in class (e.g. "how do mutexes differ from semaphores?") - Clarification of what *provided* code is doing - Understanding a compiler error - Generating additional practice examples (e.g. "can I have five examples of turnaround time calculation with FIFO and RR scheduling?")[4] - General language questions (e.g. "how do I write a do-while loop in C?")

Not allowed — feeding your assignment's specific problem statement, starter code, or exam questions to another student or an LLM for implementation help or answers: - Code and assignment answers - Exam questions and answers - Asking an LLM to write or complete a function that solves your assignment (a prompt like "write me a function to do…" is a sign you've crossed this line)

Asking an LLM for a solution to a coding assignment is conceptually the same as asking another student to write the code for you, and is treated the same way.

Use LLMs the way you'd use a TA or instructor: they're great for clarifying complex topics, offering alternative explanations, and helping you understand errors — but not for producing your submitted work. As with any outside source (including stackoverflow), critically evaluate what they tell you; they can be confidently wrong[5].

University Academic Integrity Policy

For complete academic integrity policies, please refer to the CSUMB Academic Integrity Policy.

University Policies and Resources

Enrollment and Registration

For information about requesting an incomplete or withdrawal from the course, please refer to CSUMB's Enrollment and Registration Policy.

For grade appeals, consult the Grade Appeal Policy.

Disability Services

If you have a disability that may require academic accommodations, please contact the Student Disability Resources office as soon as possible. All discussions will remain confidential. Students who have already

[4] But don't trust their math. They're getting better but not great. Practice quizzes are a better place to check this out as I try to provide walk throughs.

[5] Or, you know, come talk to me. I know a lot more about the specifics of this class than the AI does.

received approval for accommodations from SDR should schedule a meeting with the instructor as early in the semester as possible.

Collection of Student Work

Student work may be collected and used for course assessment and improvement purposes. By enrolling in this course, you consent to the collection and analysis of your academic work for educational assessment. All student work will be handled confidentially and in accordance with FERPA regulations.

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.
